Biomappings

Community Curation of Mappings between Biomedical Entities

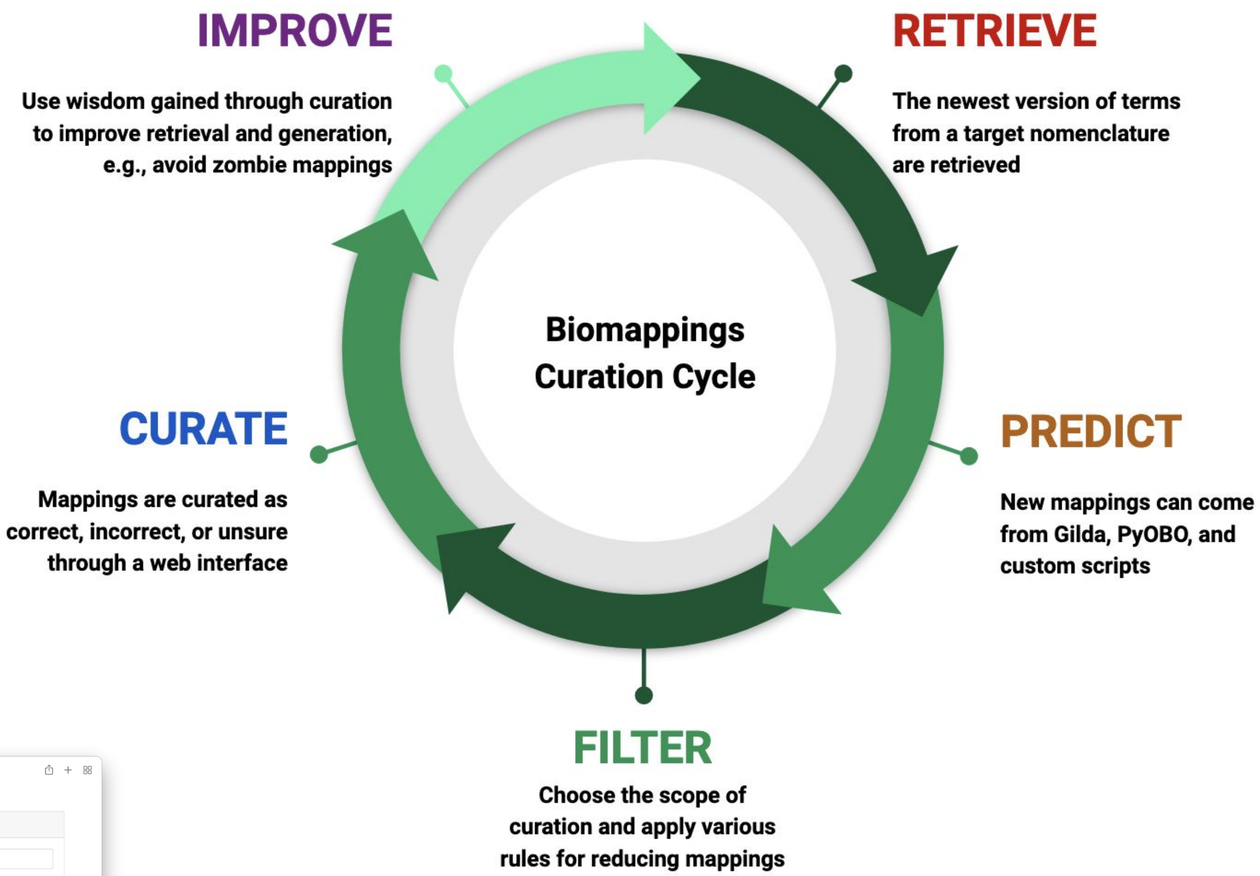
Charles Tapley Hoyt[1], Amelia L. Hoyt[2], Benjamin M. Gyori[1]

[1] Laboratory of Systems Pharmacology, Harvard Medical School, Boston, MA 02115, USA, [2] Independent researcher

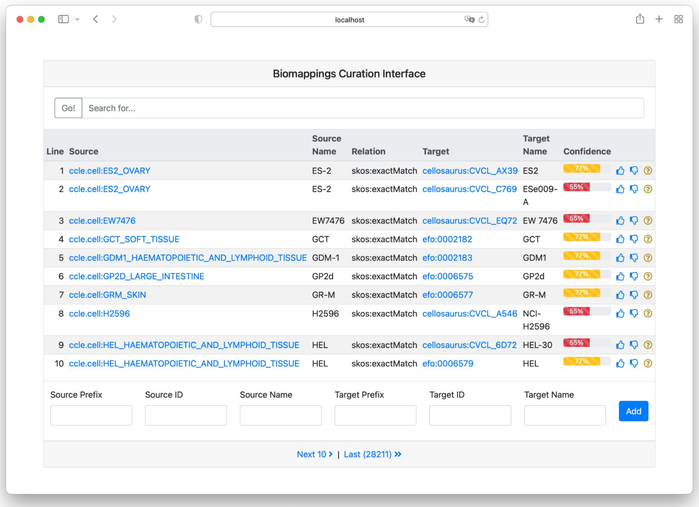
Introduction

While databases and ontologies provide some mappings (e.g., equivalences), there exist systematic gaps between widely used resources. The goals of Biomappings are to:

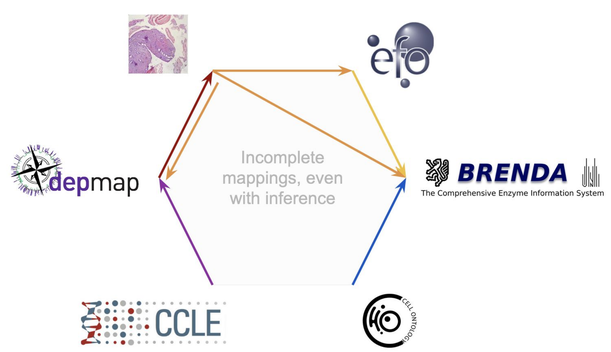
1. Implement a curation workflow to maximize coverage and curator effort
2. Curate mapping gaps based on downstream use cases
3. Couple open data with open source code for reusability
4. Encourage the inclusion of curated mappings back into primary resources



Curation Web Application



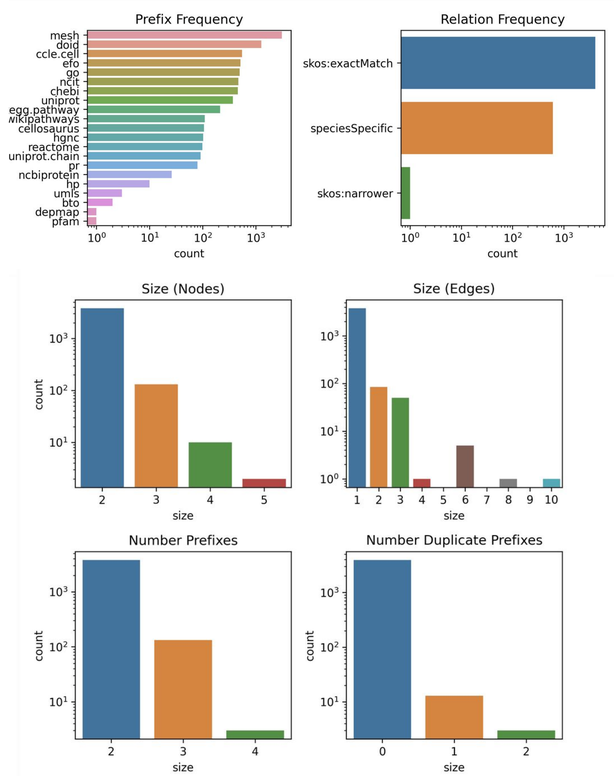
Curation of Cancer Cell Line Mappings



The mappings between cell resources covering cancer cells and cell lines are incredibly sparse. Since Cellosaurus was already the most central, we focused curation on that resource first, then the others.

Results

As of October 2021, Biomappings contains 4792 positive mappings, 910 negative mappings, 28211 predictions distributed as shown above. This includes the cancer cell line campaign, MeSH mapping campaign, pathway mapping campaign, and other smaller curation efforts.



Analysis of strongly components in induced mapping graph helps identify inconsistencies and prompt directed revisions.

Use Cases and Next Steps

Biomappings is used in the Bioontology in INDRA and now exports to the SSSOM format for easier usage in the ontology and wider mapping communities.

Source code available under the MIT License at https://github.com/biopragmatics/biomappings

Data available under the CC Zero License at https://github.com/biopragmatics/biomappings

Funded under the DARPA Young Faculty Award W911NF2010255 (PI: Benjamin M. Gyori)